Does the Universe need God?

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1. Physical cosmology does *not* provide evidence that the universe was created by God.
2. Physical cosmology does *not* provide evidence that the universe was not created by God.
3. Should I be agnostic?

A tale of two Thomases





Rev. T. Bayes

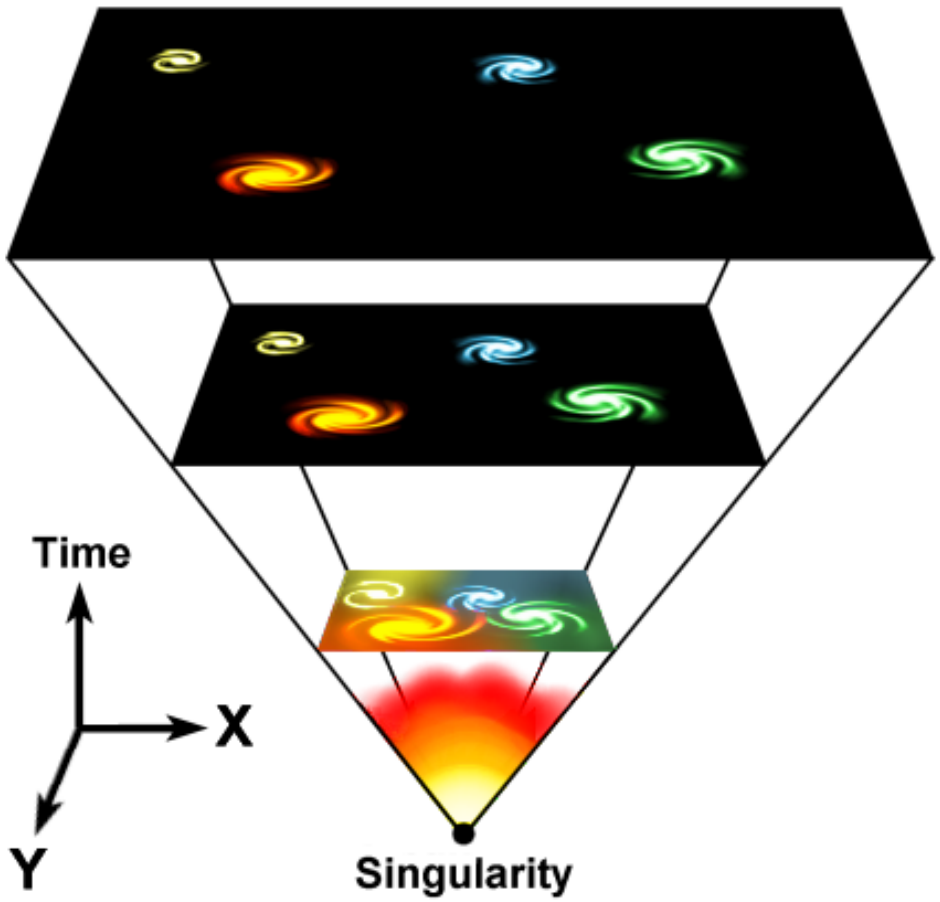
Reason can't tell us
whether or not
the universe was created

$$p(H|E) = p(E|H) \times \frac{p(H)}{p(E)}$$

PART I: Against UNG

1. SUNG: cosmology logically implies that the universe was created by God.
2. WUNG: cosmology provides evidence for the claim that the universe was created by God.

Against SUNG



Against SUNG

- No reason to trust the Principle of Sufficient Reason.
- Brian Pitts: If the initial singularity calls for a Divine Creator, then black holes call for Divine Destroyers.

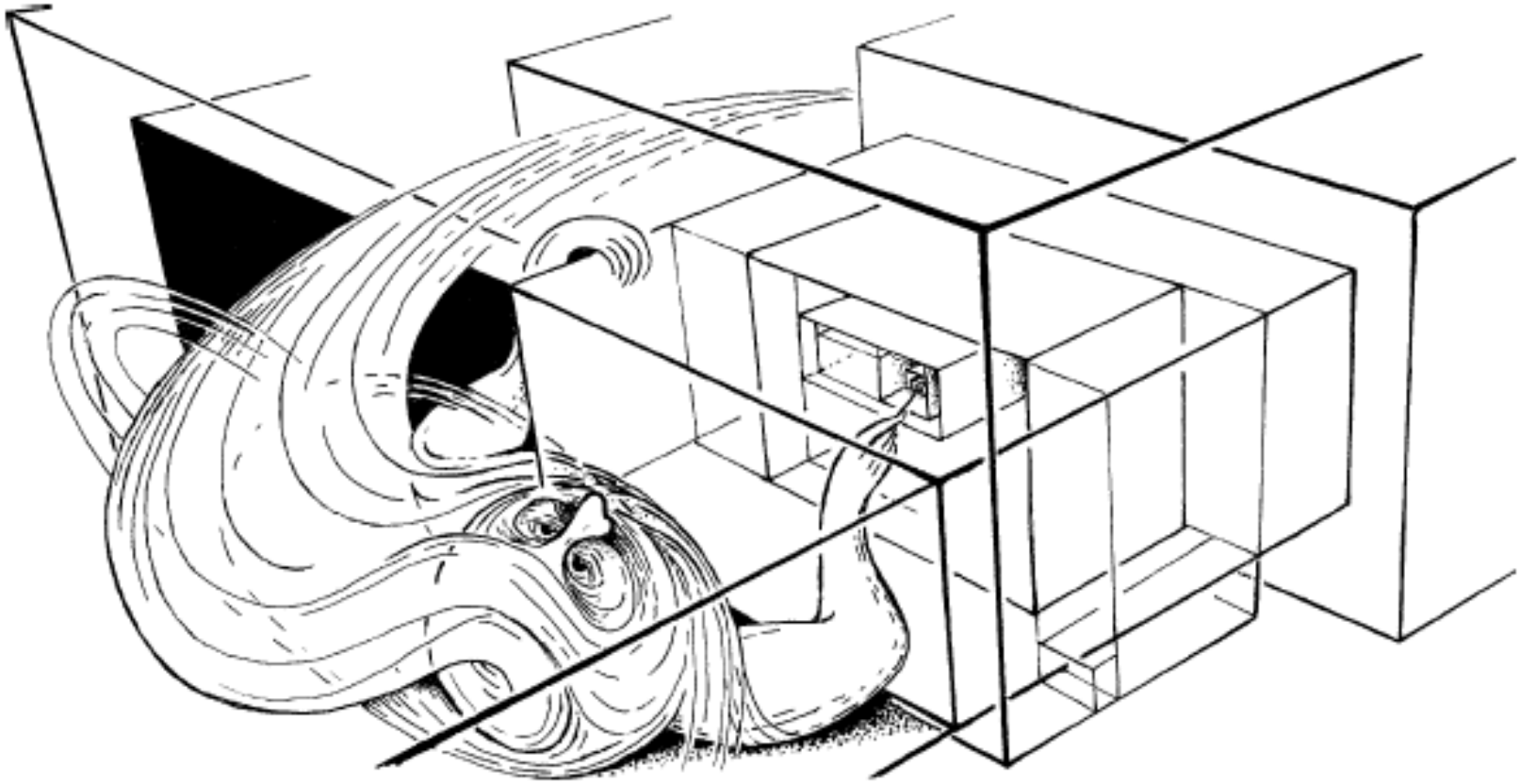
Against WUNG

A fine tuning argument

Among all of the possible universes, the "life supporting" or "biophilic" universe make up an incredibly small subset. If our universe were only a little bit different than it is, then sentient life would not have been possible. In a universe that arose by chance, most likely life would not have been possible. But life is possible in our universe. Therefore our universe did not come about by chance.

$$p(N|\neg C) \approx 0, \qquad p(N) = 1 \qquad \Longrightarrow \qquad p(C) \approx 1.$$

Against WUNG



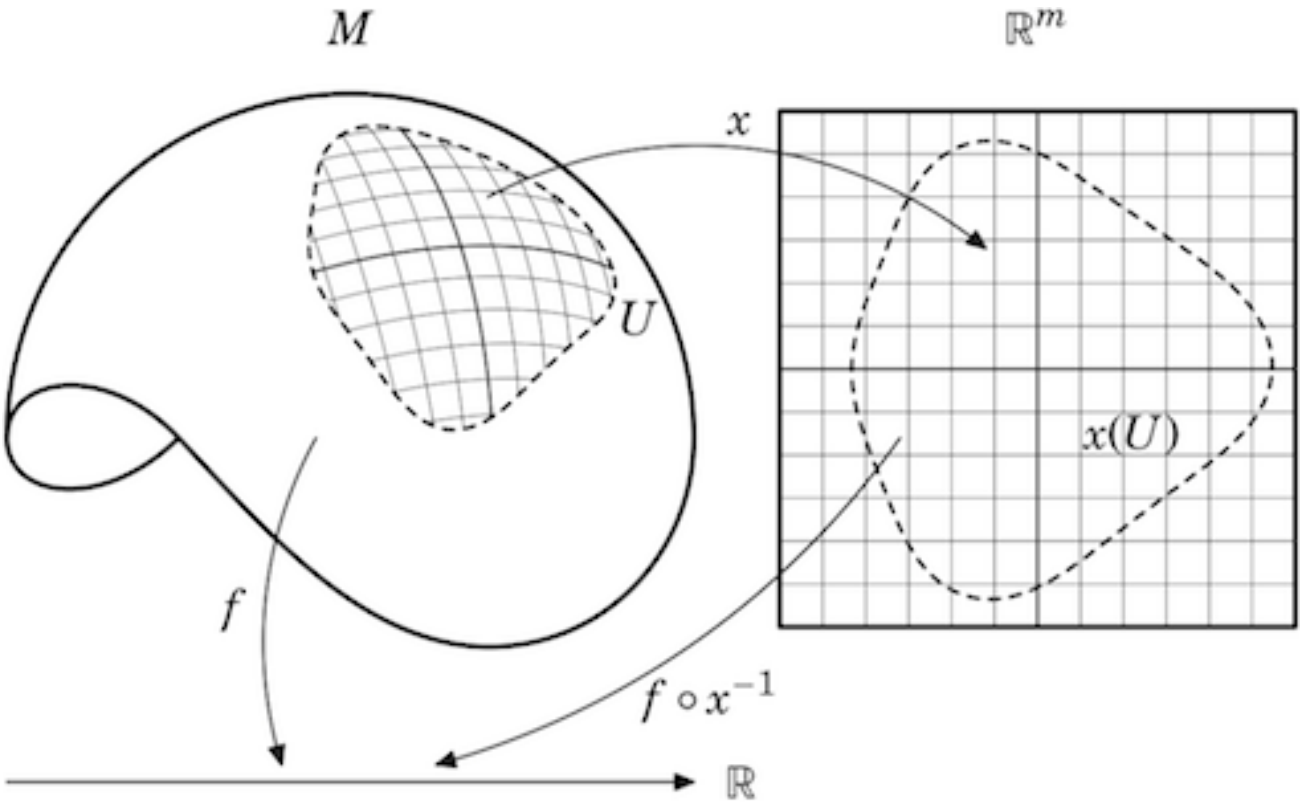
God chooses a nice universe

Against WUNG

What warrants $p(N|\neg C) \approx 0$?

1. Physics provides no probability measures over the space of values of constants.
2. GTR $\Rightarrow$ almost all universes are flat (Carroll and Tam 2011)
3. Problems with interpreting the probabilities in general relativity. (Wald 2012)

Against WUNG



Liouville measure provides a standard by which the sizes of regions can be measured.

Against WUNG

A puzzle about $p(N|\neg C) \approx 0$

- God decides which universes are possible (phase space)
- God decides which universes are likely (measure)
- The $p(-)$ of GTR is independent of C.

Interlude: Check your credence in C

- Why would the failure of these natural theology arguments reduce your credence in C, i.e. $pr(C|\neg G) < pr(C)$?
- Only if you thought that if C were true, an argument like that should be expected to work.
- But C suggests that arguments like that wouldn't work, since God wouldn't be subject to natural laws.

PART II: Against UNA

1. Physical cosmology entails that the universe was not created by God. (SUNA)
2. Physical cosmology provides evidence that the universe was not created by God. (WUNA)

Against SUNA

A caricature of Quentin Smith's cosmological argument for atheism

Premise 1: Belief that P is justified only if P figures in our best scientific explanations.

Premise 2: General relativity provides the best explanation of the origin and large-scale development of our universe.

Premise 3: General relativity doesn't say that God created the universe.

Conclusion: The claim that God created the universe is unjustified.

Against SUNA

Why might P be absent from the best scientific explanation?

1. P is false.
2. P is true, but lacks a feature we prize in good scientific explanations.
    1. P is not fundamental.
    2. P is fundamental, but lawless.

Against WUNA

A caricature of Sean Carroll's no-tuner argument

If I forget that I know anything about the universe, and if I imagine that naturalism is true, then I would expect to see something very much like what we actually find. But if I imagine that Theism is true, then I would expect to find a very different universe. Therefore, the empirical evidence confirms naturalism.

$$pr(E|\neg C) > pr(E|C)$$

Against WUNA

What warrant for $p(E|C) = \delta$?

1. Rational intuition
2. Empirical
3. Revelation

PART III: Agnosticism

1. Did *not* say that the evidence on the two sides balances out.
2. Did not recommend that you alter your credence in C, unless your credence was tethered to the facts of physical cosmology.
3. Other, non-scientific considerations are more relevant for grappling with C.
4. A scientific naturalist ought to treat C as meaningless, not as a failed attempt at a scientific explanation — for Theism takes C as a statement in metalanguage, not in the object language of natural science.

Conclusions

- Part I: cosmology based natural theology fails
- Part II: cosmology based natural atheology fails
- Part III: agnosticism not mandated by Parts I and II
- Recommendation for naturalists: remain silent about C
- Recommendation for Theists: find the truth in NOMA